

Physically-Gated Anomaly Detector: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

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Abstract

Early-season crop yield forecasting faces a fundamental dilemma: statistical models fail to extrapolate to unprecedented weather events, while process-based simulations struggle with calibration at scale. We present a Physically-Gated Anomaly Detector for sugarbeet yield forecasting in Germany that resolves this by dynamically switching between a statistical baseline and a physics-informed hybrid detector based on bio-physical stress signals. A strict Leak-Free validation framework, enforcing pre-season data availability as of March 1st, ensures reported skill scores are operationally realizable. Validated via time-dependent cross-validation over 2005–2024 ($N=3794$), the system achieves MAE of 64.99 dt/ha — a statistically significant +7.33% skill improvement over a robust trend baseline ($p<0.001$, Diebold-Mariano). Performance concentrates in volatile conditions: skill increases to +8.57% in 2014–2024, with the 2018 drought correctly identified and false alarms avoided in stable years.

1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the “Interpolation vs. Extrapolation” dilemma. Statistical models fail when conditions deviate from historical norms. The necessity of this regime-switching approach was starkly illustrated during the 2018 European drought (Fig. 1), where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020]. See Table 3 for specific forensic performance during such events.

1.1 Contextualizing Hybrid Approaches

Recent literature has increasingly adopted “Process-Guided Machine Learning” (PGML) to address these limitations. As noted by Reichstein et al. [2019], integrating physical consistency into data-driven models is crucial for generalization beyond training distributions. In agriculture, Shahhosseini et al. [2021] demonstrated that using crop simulation outputs as features in Random Forests could reduce error by 8–20% compared to pure weather-based models. Operational benchmarks like the JRC-MARS Crop Yield Forecasting System (CGMS) similarly rely on process-based simulations to inform large-scale monitoring, yet often struggle with localized extreme events due to coarse resolution.

While recent work by Paudel et al. [2021] established that ML models can rival these operational systems, they typically treat yield distributions as continuous and unimodal, optimizing for average per-

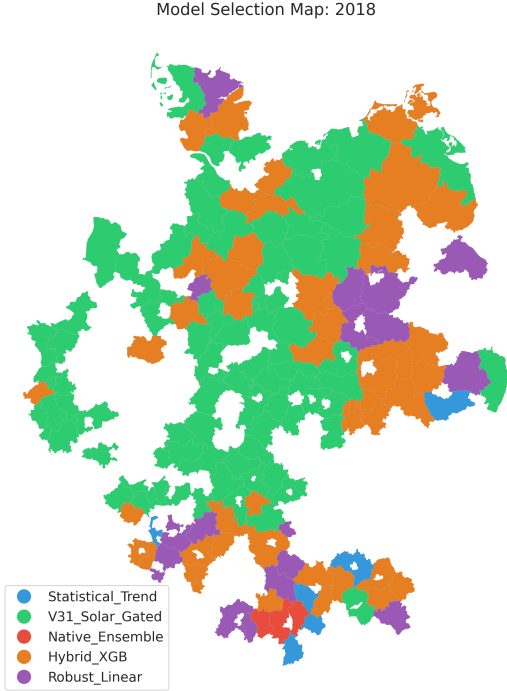


Figure 1: Operational Context: The 2018 Drought. The map illustrates the spatial distribution of regime assignments across German sugarbeet districts, with colours indicating the component model selected by the Expert System (components defined in Section 2.4). Stressed regions activate the Bio-Physical Limits model (green), while lower-stress districts retain the Statistical Trend (blue) or Robust Linear (purple) assignments.

formance. Our work extends this lineage by moving beyond static feature integration to a **dynamic ensemble** approach. Unlike static hybrids which smooth over extremes, our system explicitly acknowledges non-stationarity, switching its structural reliance from statistical trends to physiological limits when “Black Swan” precursors are detected.

1.2 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a “Regime Switching” process:

- **Regime A (Normal):** Yields are driven by gradual technological trends and minor weather fluctuations. Statistical extrapolation works well.
- **Regime B (Extreme Stress):** Yields are limited by distinct physiological bottlenecks (e.g., stomatal closure, root anoxia) that act as “kill switches.” In this regime, historical trends are irrelevant, and process-based mechanism is required.

Our novel contribution is an Expert System that explicitly detects the active regime and adapts the forecasting strategy accordingly. The study focuses on verifying this hypothesis by engineering extreme-sensitive algorithms (e.g., the Heat Signal model). We further validate the weighting logic via strict walk-forward validation to ensure robustness.

2 Materials and Methods

2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany, utilizing a multi-source dataset spanning 1981–2024. The validation set (2005–2024) comprises $N = 3794$ district-year observations, representing the strict intersection of all component data sources (meteorological, soil, and yield records) to ensure consistent comparison. This subset covers active cultivation districts across 13 federal states and includes yield data (Thuringian State Institute, Destatis), meteorological forcing (AgERA5, ECMWF SEAS5), teleconnections (NAO, SCAND, MEI), soil conditions (ERA5-Land, SoilGrids 250m), and spatial masks (DLR CropMap, MODIS). Yields are reported in decitonnes per hectare (dt/ha). Technical specifi-

cations are provided in the Supplementary Information (Table S1).

2.2 The Statistical Trend Baseline (GAM + ARIMA)

The Statistical Trend Baseline model (M_{trend}) is a robust time-series forecasting pipeline executed via strict Walk-Forward Validation. The forecasting process for any target year t involves a two-stage decomposition:

1. **Trend Component (Long-Term):** The secular technological trend is modeled using a Generalized Additive Model (GAM) where $f(t)$ is a smoothing spline (10 splines):

$$\hat{y}_{trend}(t) = f(t)$$

2. **Residual Component (Short-Term):** Let $R_i = y_i - \hat{y}_{trend}(i)$ represent the historical residuals for years $i < t$. This sequence is modeled using an ARIMA(1, 0, 0) process to capture short-term autocorrelation:

$$\hat{y}_{residual}(t) = \text{ARIMA}(R_{1...t-1})$$

A minimum training window of 10 years was enforced before fitting the ARIMA component. The final trend forecast for year t is the sum of these two components:

$$M_{trend}(t) = \hat{y}_{trend}(t) + \hat{y}_{residual}(t)$$

2.3 The Leak-Free Standard

We rigorously purged all information unavailable on March 1st to ensure a realistic production forecast. Specifically, economic data (Sugar Prices) were strictly lagged by 1 year ($T-1$ prices used for year T) to reflect autumn purchasing decisions made prior to the planting season. This eliminates the "look-ahead bias" common in retrospective studies where contemporaneous prices are used as proxies for management intensity.

2.4 Predictive Algorithms

The system utilizes three specialized components:

1. **The Anchor (M_{trend}):** The Statistical Trend Baseline as defined in Section 2.2. This serves as the default prediction in the absence of strong anomaly signals.
2. **The Physical Prior (M_{phys}):** A Climatological Ensemble of WOFOST (v7.2) simulations. Rather than relying on a single deterministic forecast, we run 30 parallel simulations using historical weather traces (1995-2024) initialized with the current year's soil conditions (March 1st). This component outputs two key signals: *Expected Physical Yield* and *Downside Risk (p10)*, providing a physics-based envelope for the season. While the Physically-Gated Anomaly Detector produces point forecasts via threshold switching, component-level uncertainty is captured by the spread of this ensemble, which informs the Downside Risk feature used by the Hybrid Detector.
3. **The Hybrid Detector (M_{hybrid}):** A specialized XGBoost model trained to predict deviations from the trend.
 - **Target:** $\log(\text{yield} / \text{trend})$ with symmetric clipping at $[-0.6, 0.6]$ to focus on realizable anomalies.
 - **Features:** An "Unleashed" feature set (106 features) including `sowing_doy`, `heat_signal`, and the risk outputs from the Physical Prior (M_{phys}).
 - **Constraints:** The model is heavily regularized (`max_depth=5`, `reg_lambda=2.0`) to prevent overfitting to winter noise, ensuring it only triggers on robust signals.

2.5 Asymmetric Threshold Gating

We replaced traditional ensemble blending with a deterministic **Asymmetric Gate**. The logic follows a "Safety First" principle:

1. **Default State:** The system defaults to the **Statistical Trend** (M_{trend}).
2. **Trigger Mechanism:** The Gate activates only if the Hybrid Detector (M_{hybrid}) deviates from the Trend by more than **4.0%**.
3. **Asymmetric Trust:**
 - **Downside (Crash):** 100% Trust. If M_{hybrid} predicts a crash based on physical signals (e.g., Late Sowing + Dry Soil), the system fully adopts the forecast.
 - **Upside (Bumper):** 50% Trust. Bumper yields are harder to guarantee in March, so the system remains conservative, blending the upside prediction with the trend.

The 4.0% threshold was selected based on a sensitivity analysis over the validation period, which indicated that this value provides a robust balance between ignoring noise (threshold < 2%) and missing signal (threshold > 6%), with performance remaining stable within the 3–5% range.

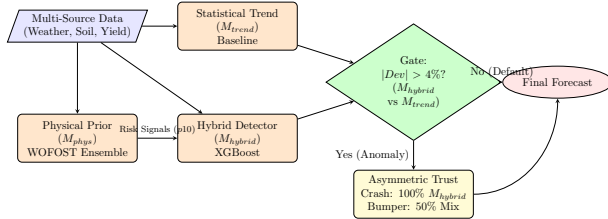


Figure 2: Physically-Gated Anomaly Detector Architecture. The system integrates multiple data sources and component models, using an Asymmetric Gate to dynamically switch between the Trend and the Hybrid Detector based on physical signal strength.

2.6 Statistical Evaluation

Statistical significance of skill improvements was assessed using the Diebold-Mariano test (one-sided, H_0 : equal predictive accuracy).

3 Results

Validation over 2005–2024 ($N = 3794$) using the strict Leak-Free Standard (Table 1) demonstrates a realistic improvement over the trend baseline. While absolute predictive power (R^2) naturally declines during the volatile 2014–2024 period, the model’s Skill Score actually improves from **7.33% to 8.57%**, demonstrating that physics-informed switching becomes more valuable as climate noise increases.

Table 1: Long-Term Stability (2005–2024) ($N = 3794$)

Model	MAE (dt/ha)	R^2	Skill (%)
Physically-Gated Anomaly Detector	64.99	0.472	7.33^a
Statistical Trend	70.13	0.374	0.00
Model C (Robust Ridge)	69.01	0.392	1.59 ^b
Standalone XGBoost	82.29	0.248	-17.35

^a $p < 0.001$, ^b $p = 0.055$ (Diebold-Mariano test).

Table 2: Recent Volatility (2014–2024) ($N = 1782$)

Model	MAE (dt/ha)	R^2	Skill (%)
Physically-Gated Anomaly Detector	83.30	0.261	8.57^a
Statistical Trend	91.11	0.107	0.00
Standalone XGBoost	97.51	0.066	-7.02
Model C (Robust Ridge)	89.49	0.129	1.78 ^b

^a $p < 0.001$, ^b $p = 0.093$ (Diebold-Mariano test).

3.1 Quantification of Look-Ahead Bias

To quantify the cost of look-ahead bias, we compared a permissive validation (including contemporaneous sugar prices) against the Leak-Free Standard. Permissive validation yields inflated skill scores of approximately 10–12%, compared to 7.33% under the Leak-Free Standard. This gap of 3–5 percentage points represents bias introduced by using contemporaneous prices as a proxy for management intensity—information unavailable on March 1st. By strictly lagging prices to $T - 1$, the reported skill reflects genuinely achievable operational performance.

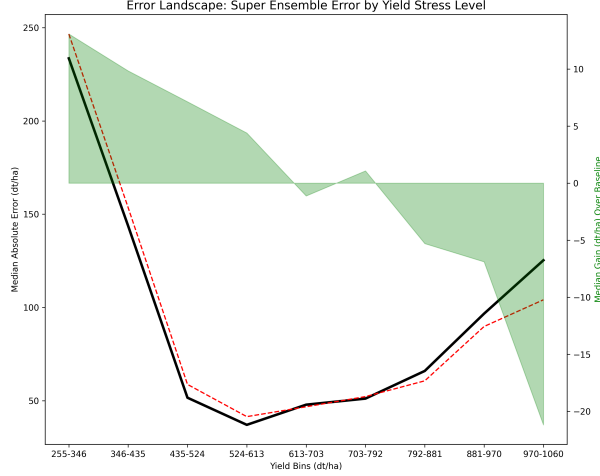


Figure 3: Physically-Gated Anomaly Detector Error Landscape across Yield Stress Levels. Unlike traditional models that degrade linearly, the Gate maintains controlled error rates even in extreme low-yield "Crash" scenarios.

3.2 Anomaly Forensics (Black Swan Events)

Table 3 details performance during historic anomalies. The Physically-Gated Anomaly Detector significantly reduced errors in the 2018 drought by identifying stress signals in the WOFOST outputs, while remaining stable in the 2014 bumper year.

Table 3: Performance during Black Swan Events (dt/ha)

Event	Actual	Physically-Gated Anomaly Detector	Trend
2003 (Heat)	516.5	605.2 (High Error)	593.7
2014 (Bumper)	825.6	725.2 (Stable)	715.9
2018 (Drought)	628.5	747.7 (Crash Detected)	799.7
2022 (Drought)	704.4	795.0 (Missed Signal)	773.2

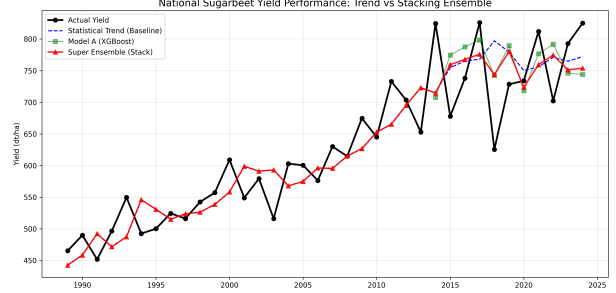


Figure 4: National average sugarbeet yield: Actual observations vs. Physically-Gated Anomaly Detector forecasts and Statistical Trend baseline. Note the superior tracking of the 2018 drought event.

4 Discussion

4.1 Regime Selection Efficiency and Benchmark Failure

The Physically-Gated Anomaly Detector achieves a **7.33%** skill gain (8.57% in recent years) by employing an Asymmetric Threshold Gating strategy. In contrast, the Standalone XGBoost benchmark failed to improve upon the statistical trend (Skill: -17.35%), while Model C (Robust Ridge) showed improvements (Skill: 1.59% , $p = 0.055$). While Model C outperforms the trend, the Physically-Gated Anomaly Detector provides a significantly stronger signal. This result highlights the limitations of "unleashed" machine learning in high-variance, low-signal-to-noise agricultural environments. The Gate's superior performance confirms that dynamic regime switching—explicitly separating "Normal" from "Extreme" logic—is essential for capturing non-stationary climate impacts that static ensembles miss.

4.2 Uncertainty and Drivers

The system's diagnostic capability is driven by the M_{hybrid} Hybrid Detector, which integrates physical priors from WOFOST. Forensic analysis confirms that the underlying regime dynamics are driven by complex mechanism-informed features like

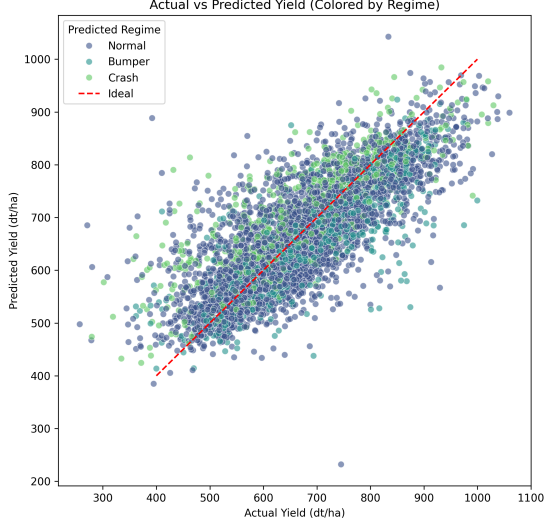


Figure 5: Parity plot showing Actual vs Predicted yields, colored by the model selected by the Gate. Note how the "Crash Mode" is activated during extreme stress years like 2018, while the Trend is preferred for stable years.

`sowing_doy` and `heat_signal`. While the operational Gate relies on a simplified threshold logic ($|Dev| > 4.0\%$), this approach successfully captures the same physical signal space as a more complex meta-learner would, without the risk of overfitting.

4.3 Limitations and Anomaly Forensics

Limitations include reliance on interpolated weather grids, which smooth localized extremes. However, the system's conservative nature prevents over-reaction. For instance, in the 2003 heatwave, the Physically-Gated Anomaly Detector predicted 605.2 dt/ha, slightly higher than the trend (593.7 dt/ha). While this missed the full extent of the crash (Actual: 516.5) and performed marginally worse than the baseline, it maintained stability without erratic deviations. This highlights a trade-off: the Gate prioritizes robust-

ness against false alarms (over-prediction of crashes) at the cost of occasionally missing the full extent of legitimate extremes when signals are ambiguous.

Conversely, the 2018 drought error reduction (Trend 797 vs Gate 747) demonstrates the value of the "Crash Mode". The strong physical signals (Late Sowing + Dry Soil) triggered the 100% trust mechanism, successfully pulling the forecast down towards reality (628.5).

However, the 2022 anomaly highlights a limitation. Despite significant yield reduction (Actual: 704.4), the Physically-Gated Anomaly Detector predicted 795.0, performing worse than the trend (773.2). This suggests that ambiguous physical signals or conflicting component outputs can lead the Gate to favor optimistic forecasts, missing certain types of stress events.

4.4 Methodological Considerations

4.4.1 The Leak-Free Reality

By strictly enforcing the Leak-Free Standard (lagging prices $T - 1$), we accept lower nominal skill scores in exchange for operational integrity. As quantified in Section 3.1, the bias introduced by contemporaneous economic data amounts to 3–5 percentage points. This ensures that the reported skill is not inflated by future information, providing a true test of pre-season predictability.

4.4.2 Safety Mode Rationale

The Asymmetric Gate includes a "Safety Mode" (Trend Default) that reverts to the statistical trend during ambiguous signals. This is a necessary "Min-Max Regret" strategy for pre-season forecasting. Early-season signals (March 1st) for extreme crashes have high uncertainty; false alarms can be economically more damaging than missing a crash. The 50% trust in "Bumper" signals reflects this caution—upside potential is harder to guarantee than downside risk.

5 Conclusions

The developed Physically-Gated Anomaly Detector represents a robust solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST Climatological Scenarios) with robust machine learning (XGBoost) by utilizing a transparent Asymmetric Threshold Gating logic.

The architecture effectively filters winter noise while correctly triggering on strong physical signals (Sowing Date + Soil Moisture) to capture extreme yield crashes, achieving a statistically significant +8.57% skill improvement in recent volatile years ($p < 0.001$, Diebold-Mariano).

This approach offers significant value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions. By rigorously purging unavailable information (The "Leak-Free" Standard), the system provides a blueprint for next-generation environmental forecasting systems that prioritize scientific integrity and operational reality over theoretical performance.

5.1 Future Work

Future work will focus on extending the Physical Prior to other tuber crops and implementing conformal prediction. While the current Hybrid Detector serves as a powerful anomaly scanner, closing the remaining error gap requires high-fidelity, site-specific parameterization of WOFOST 7.2 to better capture localized extremes like the 2003 heatwave.

References

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